UCC 103 - PRINCIPLES OF COMPUTING

Chapter 12: Database System Application

At the end of the chapter the learner shall be able to;

- Explain the different database terms
- Open and use an existing database
- Design and create a simple database
- Save database objects with appropriate names
- Create simple queries
- Design input forms
- Design output reports

12.1. Introduction

Database Definition:

- *A shared collection of logically related data, and a description of this data, designed to meet the information needs of an organization.*
- *a mechanized formally defined centrally controlled integrated collection of logically organized data in an organization*

Data are known facts that can be recorded and that have implicit meaning.

A database has the following implicit properties;

1) A database represents some aspect of the real world called **mini-world** and changes in the real world are reflected in the database
2) A database is a logically coherent collection of data with some inherent meaning
3) A database is designed built and populated with data for a specific purpose.
4) A database has some source from which data is derived some degree of interaction with events in the world and an audience that is actively interested in its contents.
5) It has an intended group of users and some preconceived applications in which these users are interested.
6) A database can be of any size and complexity and may be generated and maintained manually or it may be computerised.

12.2. Database Management System (DBMS)

1) A database management system (DBMS**)** is a collection of programs that enables users to create and maintain a database.
2) A software system that enables users to define, create, maintain, and control access to the database.

The DBMS is a general-purpose software system that facilitates the following processes among various users and application;

1) Defining database: It allows users to define the database, usually through a **Data Definition Language** (DDL). The DDL allows users to specify the data types structure and constraints of the data to be stored in the database..

2) Constructing database: the process of storing the data on some storage medium that is controlled by the DBMS.
3) Manipulating database: It allows users to insert, update, delete, and retrieve data from the database, usually through a **Data Manipulation Language** (DML)
4) Sharing database: is the process of allowing multiple users to access the database simultaneously
5) Data Security - It provides controlled access to the database. For example, it may provide:
   a) a security system, which prevents unauthorized users accessing the database;
   b) an integrity system, which maintains the consistency of stored data;
   c) a concurrency control system, which allows shared access of the database;
   d) a recovery control system, which restores the database to a previous consistent state following a hardware or software failure;
   e) a user-accessible catalog, which contains descriptions of the data in the database.

Other DBMS also provides protection to the database which includes system protection against hardware and software security protection against unauthorised or malicious access. It also facilitates presentation and Visualization of data as well as maintain and the database and associated programs over the lifetime of the database application

Examples of DBMS are Ms Access, Paradox, Dbase, FoxPro, Sybase.
From the above examples, we shall look at Ms Access.

12.3. Advantages of a Database Management System

Benefits of using database to store information
1) Fast retrieval of information i.e. data is easier to find.
2) Easy to maintain accurate and up to date data.
3) Easy to analyze and make summary reports on the stored data.
4) Easy to protect your data from unauthorized access.
5) Information can be accessed in many ways.

12.4. MS ACCESS 2007/2010

Microsoft Access 2010 is a powerful relational database program that includes hundreds of tools you can use to quickly start tracking, sharing, and reporting information, even if you are new to database development. Users have access to a large library of professionally designed templates; wizards that automatically create tables, forms, queries, and reports; and extensive local and online help resources.

Microsoft Access is a relational database package that runs under Windows on microcomputers. It enables you to:-
1) Build a database - Plan and create a database; create a table; add fields and data to the table.
2) Manage data. Sort records; use filters to sort data; create and run a query.
3) Analyze data. Create and use forms to access and analyze information in

12.4.1. Structure of MS ACCESS DBMS

There are seven main elements of a database in Access:

1) Tables. Use tables to store data.
2) Queries. Filter data so that you retrieve selected records or fields from the database.
3) Forms. Use forms to enter or edit the data in your tables. Forms let you view one record at a time.
4) Reports. Use reports to deliver a professional presentation or written report.
5) Modules. Automate a group of related procedures in Access.

Others Include

6) Data access pages - Hyperlinks. Create HTML pages from a database quickly and efficiently.
7) Macros. Automate tasks that you perform on a regular basis in a database.

1. Table:

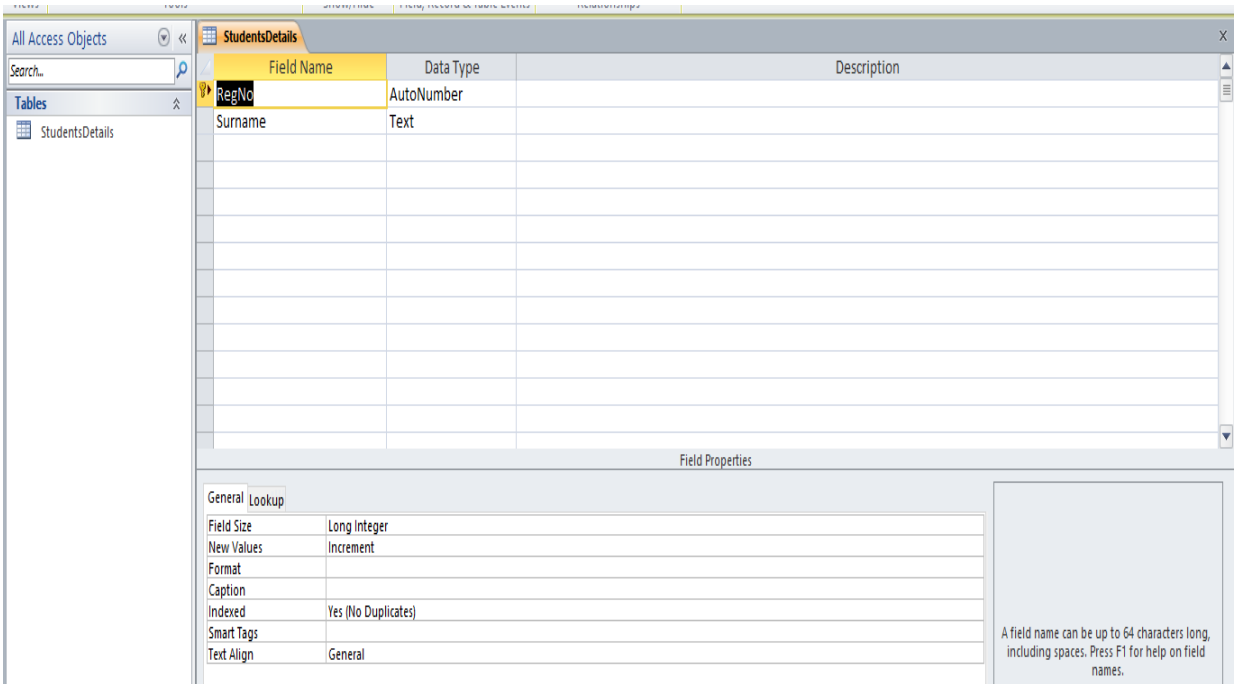
   Collection of data about a specific subject e.g. customers, employees e.t.c. In a table, data is organized in fields (columns) and records (rows).

   a) ***Field:*** This is any piece of information in a database e.g. Names, Age, Birth date.
   b) ***Record:*** It is a collection of several related fields.

   Structure of the Table

   a) ***Field name***: - these are column headings for the table being created.
   b) ***Data types:*** - the attribute of a field that determines the type of data it can hold e.g. Text, Number, Currency, Date & Time, Yes/No, Lookup & Relationship, Rich Text, Memo, Attachment, Hyperlink, Calculated Field etc.
   c) ***Description:*** - This column is optional but can be used to give a detailed of the field name.
   d) ***Primary key***: - This is a field that is used to uniquely identify each record stored in a table.

Table under design View



Data Types allowed

Data type	Use	Example
Text	(Default) text or combination of text & numbers as well as numbers that don't require calculations such as phone numbers. The default size is 50 characters but up to 225 characters can be stored.	Stephen, Mary, KAZ 078w KQ2534
Memo	Lengthy text or combination of text and numbers. Can store up to 64,000 characters	Notes and descriptions
Number	Numerical data used in mathematical calculations, except that involving money.	6634, 76731, 890987, 33,978, 100.09
Date and time	Dates and times values for the years 100 through 9999	2-jan,17/8/1990, 09:55AM 4-march-1989
Currency	Monetary values	$5689, Ksh9873, £5637
Auto number	A unique sequential (incremented by 1) number or random number assigned by Ms Access whenever a record is added. AutoNumber cannot be updated	
Yes/No	Yes or No values or fields contains only one or two values (true/false, yes /no, on off)	

2. **Queries**
   A query is a type of database search. It enables you to retrieve data that has met conditions you specify, indicating data from tables. The word 'query' literally means 'to ask'. Access queries provide a way of asking about your data. When you design a query, you identify the fields to be included and the records to be retrieved from one or more tables. You can as well update or delete multiple records at the same time, perform built in or custom calculations on your data.
3. **Forms**
   They are used for adding new data, editing or displaying existing data. This data can be in a table or a query. Information entered directly in a form will appear in Tables automatically.
4. **Reports**
   The desired result of any database is to provide information in the form of reports. You can print reports from tables or queries in any desired format.
   Reports are used to summarize and display data from your database. It displays the most up-to-date information.
5. **Modules**
   They are collection of codes using Access Basic programming language. If programming is necessary, you can write modules.

12.4.2. Applications of databases

1) *Educational applications:* Preparing students' reports, class schedules.
2) *Retail applications:* This may include sales projections and Quotas, Market analysis e.t.c.
3) *Legal applications:* This may include client information and case schedules.

4) *Financial applications:* This may include stock market forecasting and end of year financial reports.

12.4.3. Planning requirements

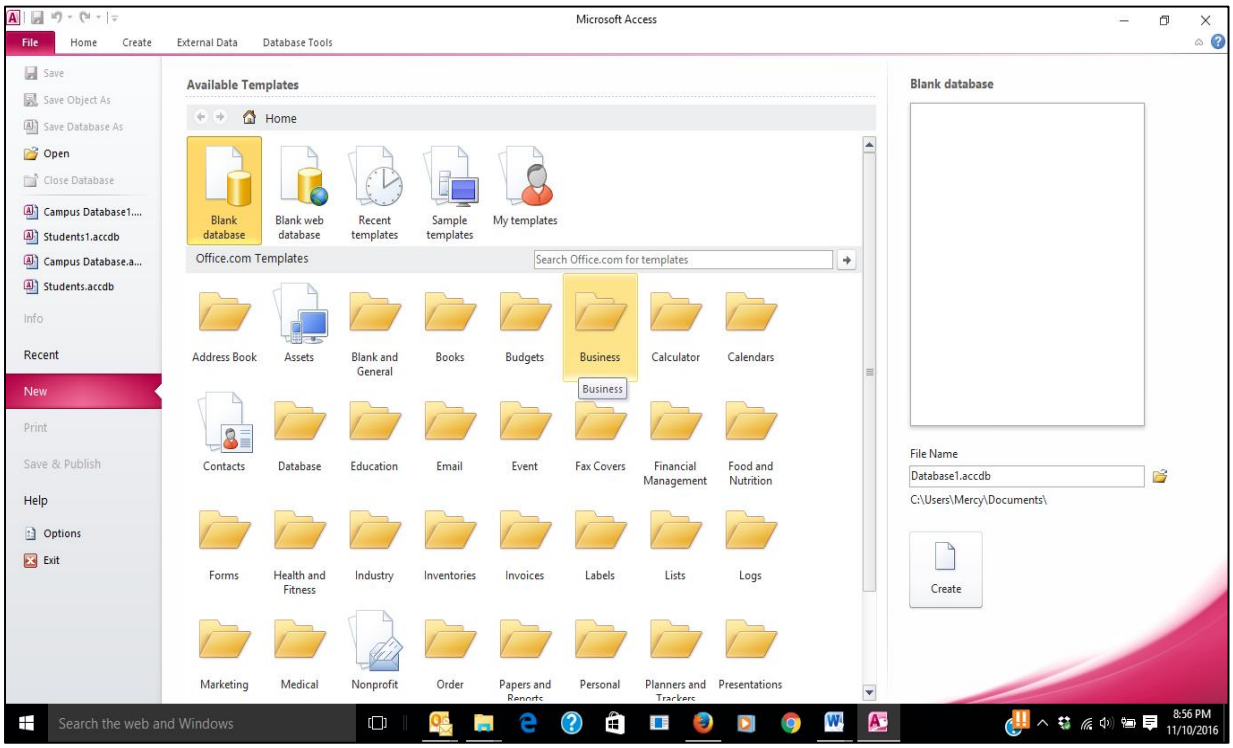
In order to plan an efficient database you need to ask yourself some questions, these include.

1) What information do I need to keep track on?
2) What is my hardware and software capability?
3) Who will be using the database?
4) What are the projected future database needs?

12.5.WORKING WITH ACCESS 2010

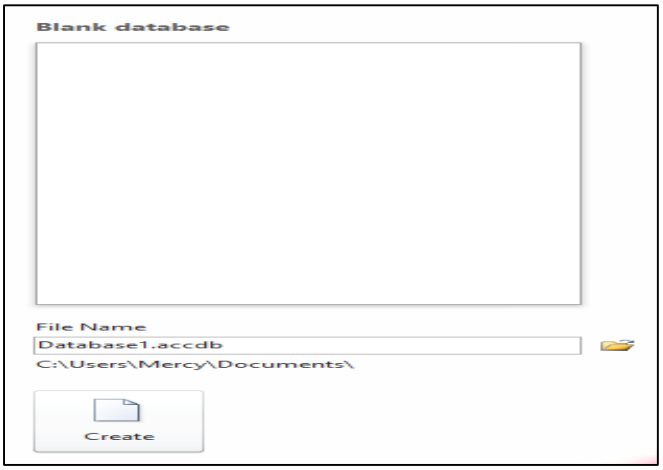
Starting Access

1. Open MS Access from the Microsoft Office Program Folder.
2. The resulting dialog box prompts you to create a new database using either blank database, database wizard or existing database.

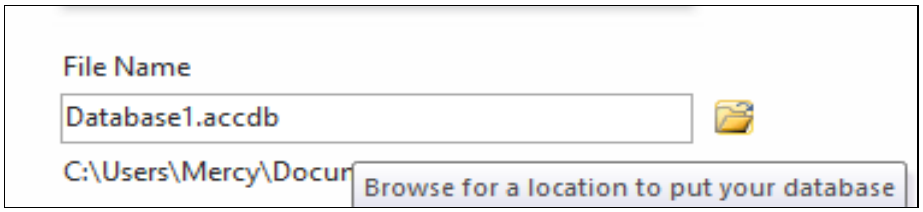


3. Click ‘blank database, option since you are creating a new one.
4. Enter the File name for Database been created. Example Students Academic Records
   File name extension is **.accdb**

NB. Select an appropriate Name for the Database based on the type of information been stored, managed and manipulated



To Change the storage location

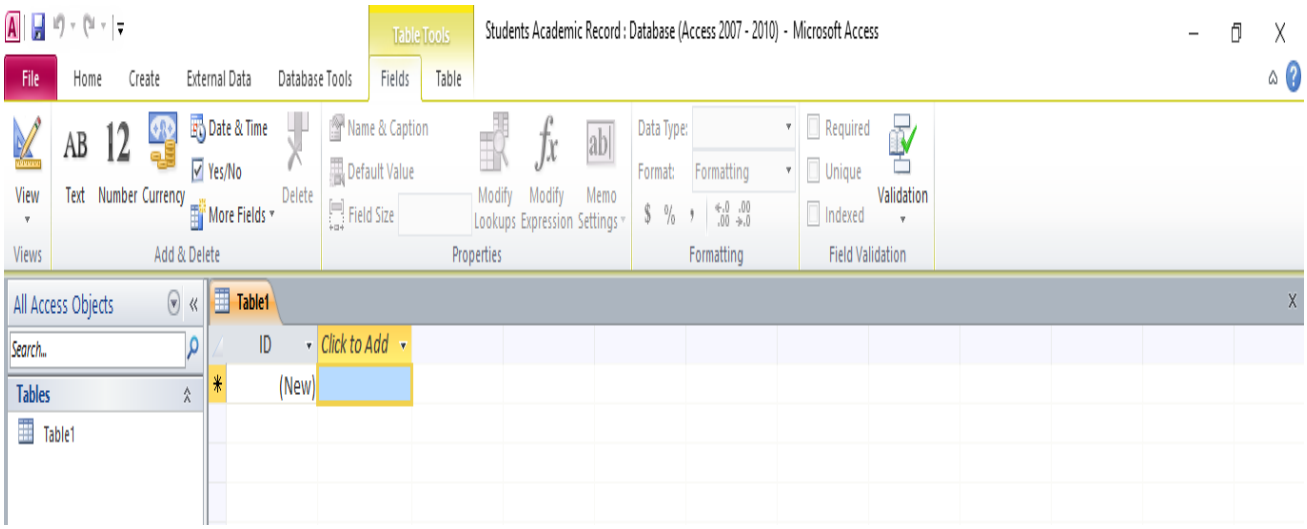


5. Click Create button

12.5.1. MS ACCESS Environment

NB: The database name ***(Students Academic Record: Database)*** is displayed on the title bar of the database window.

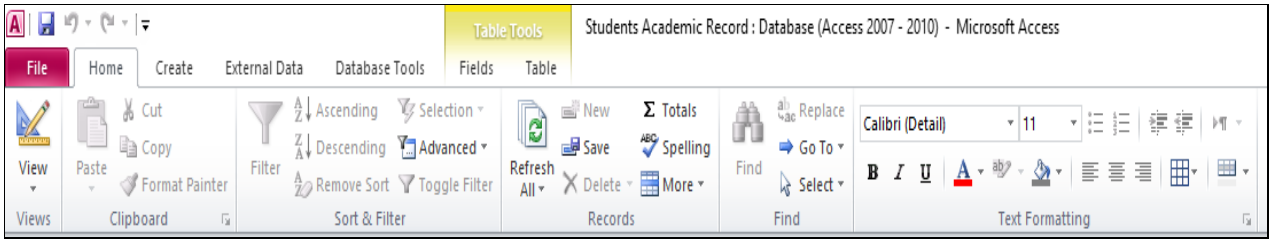
Window opens with the Fields Menu selected; - in preparation for creating the Database records starting by defining the Columns/ fields



On the left side, the list of Database tables is displayed.

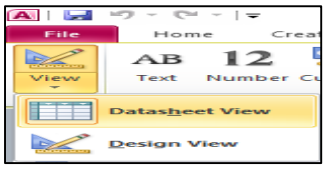
12.5.2. Menus

Home Menu



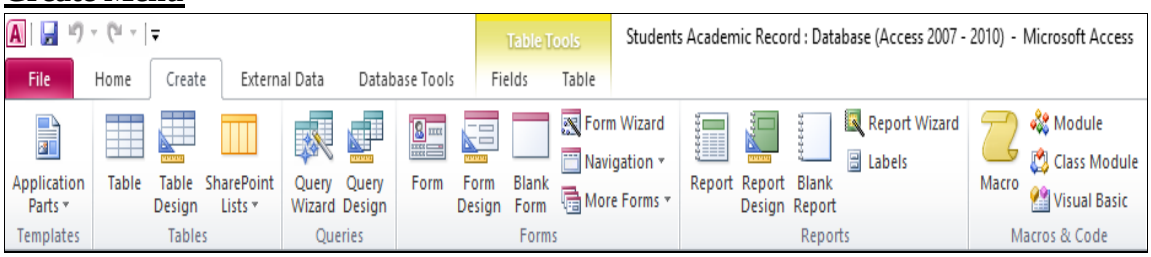
Sub-Menus

1) View: provide different view options – 2 main views



2) Clipboard
3) Sort & Filter
4) Records
5) Find
6) Text Formatting

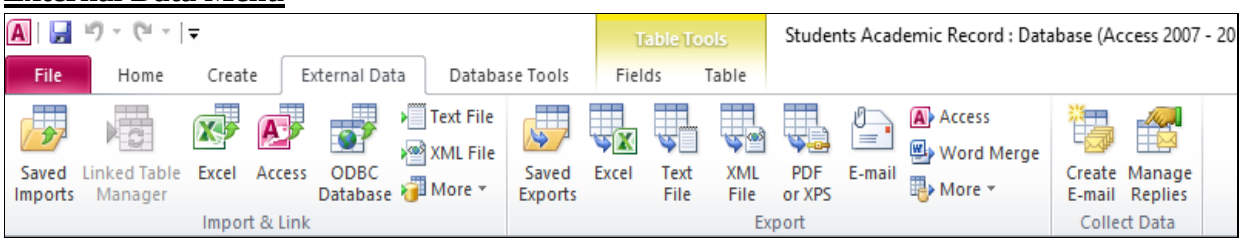
Create Menu



Sub-menus

1) Templates
2) Tables
3) Queries
4) Forms
5) Reports
6) Macros & Code

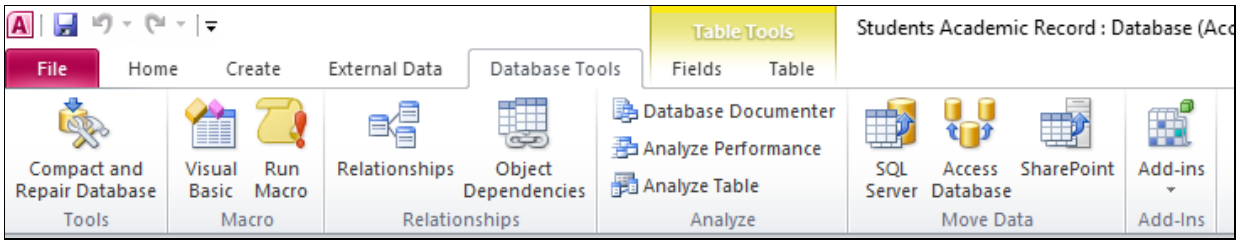
External Data Menu



Sub-Menus

1) Import & Link
2) Export
3) Collect Data

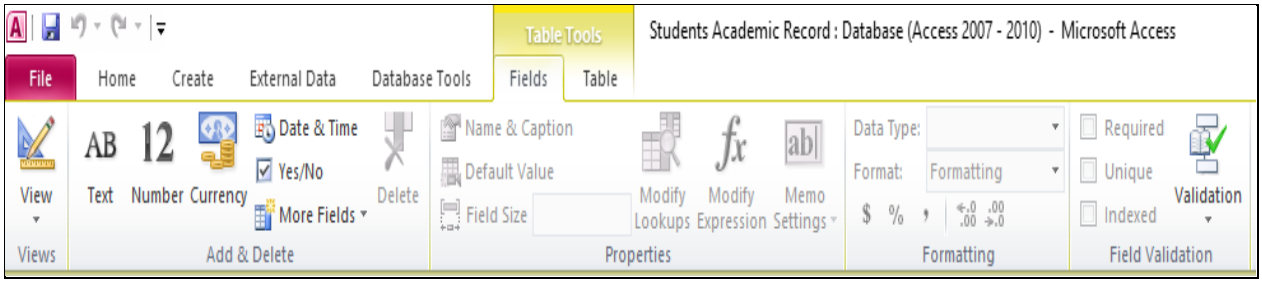
Database Tools Menu



Sub-Menus

1) Tools
2) Macro
3) Relationships
4) Analyze
5) Move Data
6) Add-ins

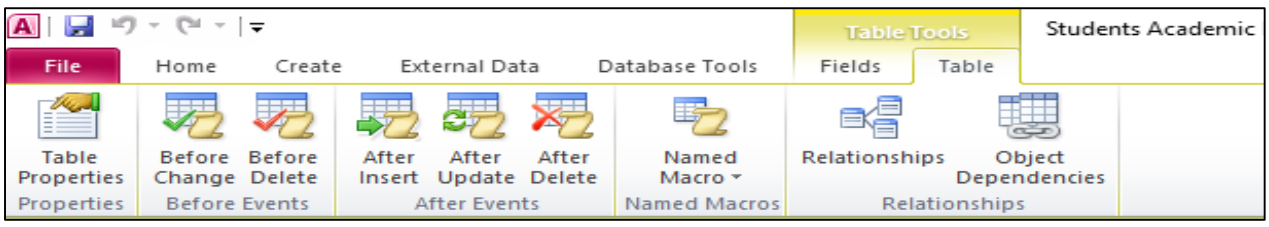
Fields Menu



Sub-Menus

1) Views
2) Add & Delete
3) Properties
4) Formatting
5) Field Validation

Tables Menu



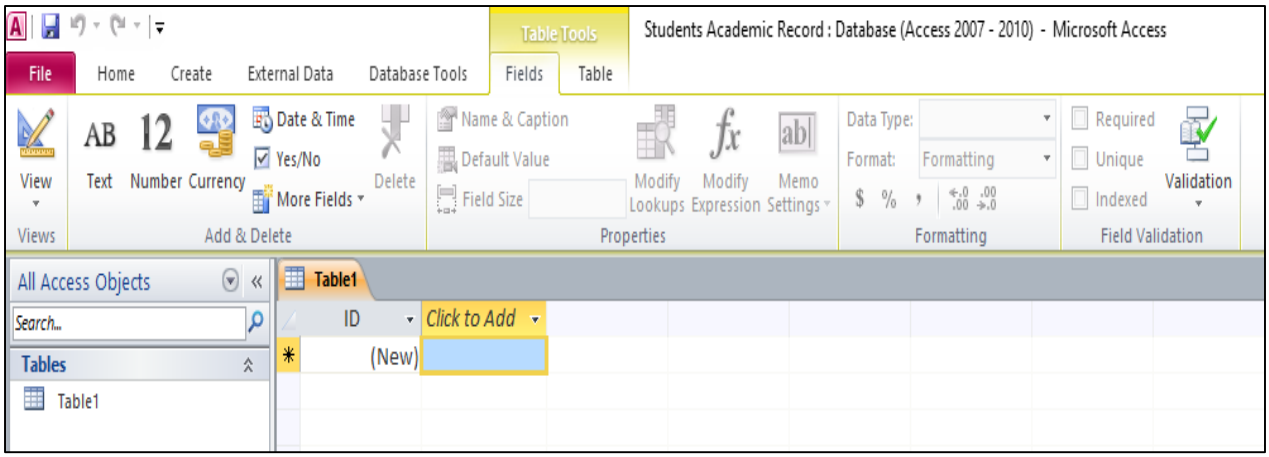
Sub-Menus

1) Properties
2) Before Events
3) After Events
4) Names Macros
5) Relationships

12.6.Working with Tables

When you click create new database, Access opens the following screen to allow you starting defining the database tables.

Note the Datasheet menu is activated.



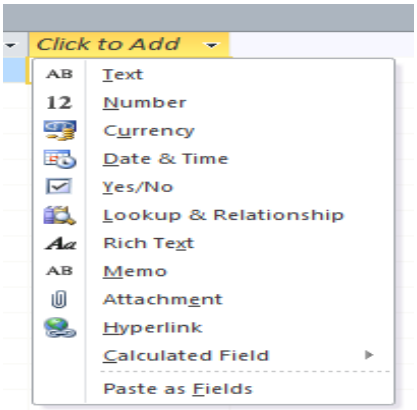
Two default, ID and Click to Add Fields are created.

To Change Field name

1) Select the field, and Right Click
2) Select rename
3) Enter the new name.

To Add new fields

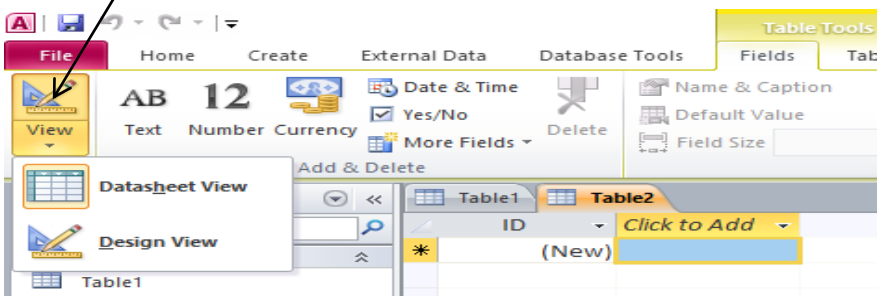
1) Select/ click the “Click to Add” field
2) Select the Data type for the field



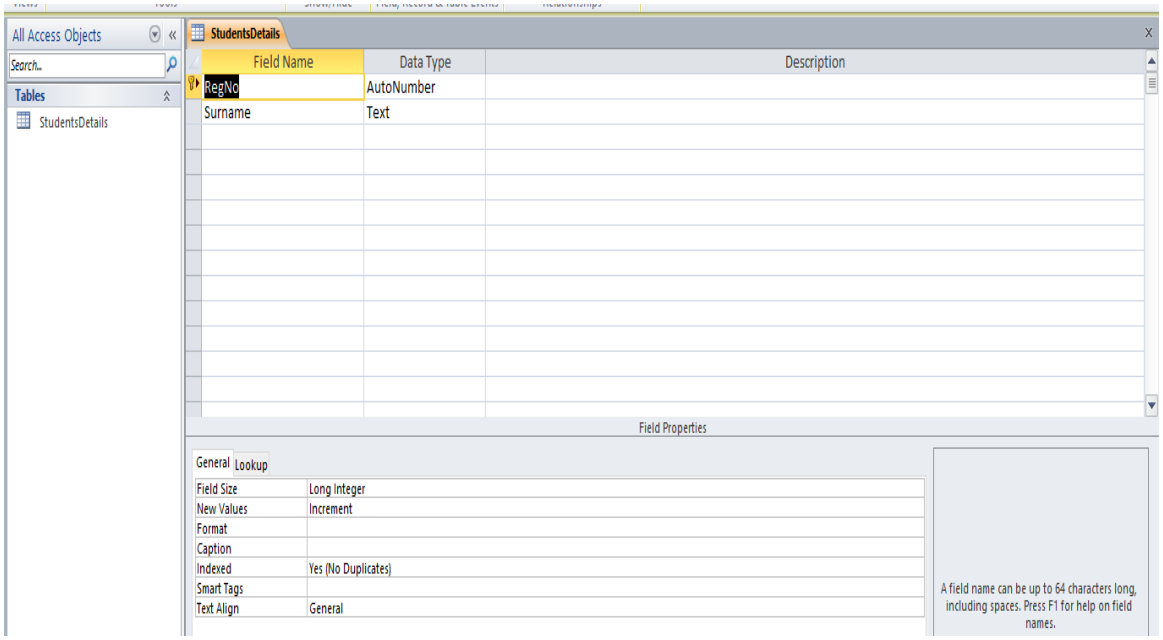
3) Select the Field again and Right Click
4) Select Rename and type in the corresponding name
5) Repeat the above steps till all the required fields have been created
6) Save the Table

Using Design View

An Alternative method for table Fields creation is to use the Design View
Go to Home menu
Click Views sub-menu



Select Design View



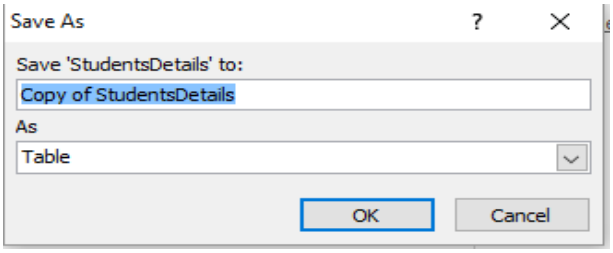
1) Enter the Field Name
2) Define the Datatype for the field
3) Type a description about the field
4) Save the Table

NB. When saving the table for the First time, Access required you to provide a Table Name. Choose an appropriate Table name based on the data/details been stored

Changing Table name.

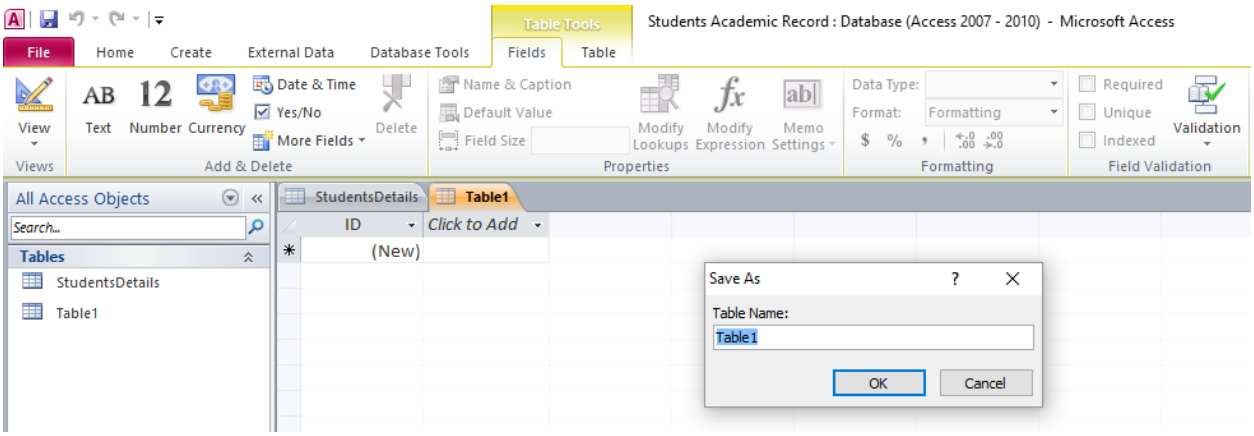
With the respective table selected/opened

1) Select File Menu
2) Select Save Object As
3) Enter the New Table name
4) Select Table in the **As** (type) section since the Object is a Table
   (other Database objects – Form, Queries, Reports and Modules)
5) Click OK button or Cancel to abandon the procedure



12.7.To Insert / Create a New Table in an Existing Database

1) On the Create Menu
2) Select Table
3) A new table (Table1 or Table n) name appears among the List of Database Tables



4) Define the Table fields are required
5) Save the Table to give it a Name

NB . By Changing the View from Datasheet to Design View, before a table is saved, MS Access automatically open the Save As Form requiring the table name to be provided first.

Table Primay Key

A unique field is required in every table. This is an identification key like the national ID card that identifies each person individually. The unique key is referred to as **primary key** in Access. It is useful when you want to avoid entering duplicated records.

The first field name is by default assigned Primary key status



12.8.Creating a Database for Students Academic records

The subject/object here is a student and the suggested field names could be;
Planning stage

1) Identify the possible details to keep
2) Define the tables (containers) and possible fields

Create the following Tables;

Table Name: Students

Field Name	**Data Type**	
StudentReg_No	Autonumber	Primary Key
First_Name	Text	
Surname	Text	
Address1	Text	
Address2	Text	
Address3	Text	
Email	Text	
Cellphone	Number	
DateofBirth	Date/time	
Sex	Y/N	
CourseID	Number	
Admission Date	Date/Time	

Table Name: Courses

Field Name	**Data Type**	
CourseID	Autonumber	Primary Key
CourseName	Text	
CourseDuration_yrs	Number	
Max Units	Number	
Min Units	Number	

Table Name: Units

Field Name	**Data Type**	
UnitID	Autonumber	Primary Key
UnitName	Text	
CourseID	Number	
ContactHours	Number	
Pre-requisite_UnitID	Number	

Academic Year

Field Name	Data Type	
AcYearID	Autonumber	Primary Key
AcYearName	Text	
AcYear_Startdate	Date	
AcYear_Enddate	Date	

Academic Semesters

Field Name	Data Type	
AcSemID	Autonumber	Primary Key
AcSemName	Text	
AcSem_Startdate	Date	
AcSem_Enddate	Date	
AcYearID	Number	

Table Students_Units_Registration

Field Name	Data Type	
UnitRegID	Autonumber	Primary Key
UnitID	Number	
StudentRegno	Number	
AcSem_Code	Number	
DateUnitRegistered	Date	
Marks	Number	
Grade	Text	

12.9. Relationships

After setting up different tables for each subject in a database, there is need to let access know how to bring the information back together. The first step in this process is to define relationships between the tables. After that, queries, forms, reports can be used to display the information from all the tables at once.

A relationship is an association established between common fields in two tables. It can be

1) one-to-one, 1:1
2) one-to-many 1:M
3) many-to-many `M:M

A relationship works by matching data in key fields usually a field with the same name in both tables.

A one-to-one relationship

In a one-to-one relationship, each record in Table A can only have one matching record in Table B and each record in Table B can have only one matching record in Table B.

A one-to-many relationship

A one-to-many relationship is the most common type of relationship. In a one-to-many relationship, a record in Table A can have many matching records in Table B, but a record in Table B has only one matching record in Table A.

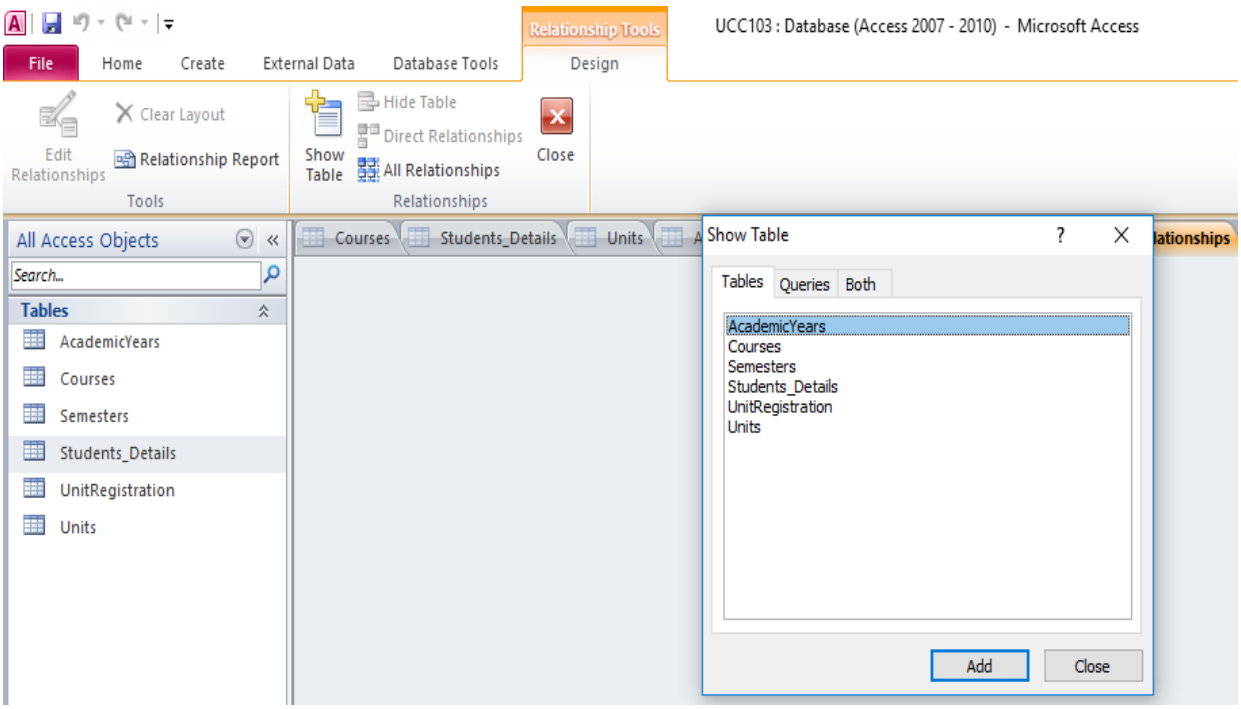
A many-to-many relationship

In a many-to-many relationship, a record in Table A can have many matching records in Table B, and a record in Table B can have many matching records in Table A.

12.9.1. Creating a Relationship

Related fields **MUST** be of the same size and data type

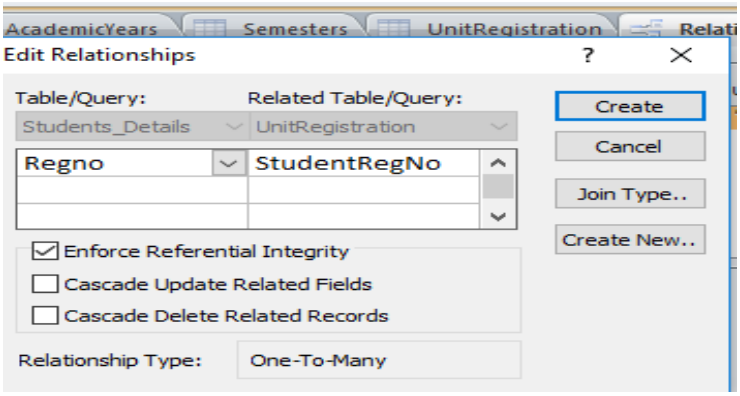
1) Select the Database Toolsn menu
2) Select Relationships sub-menu – to Open show tables window



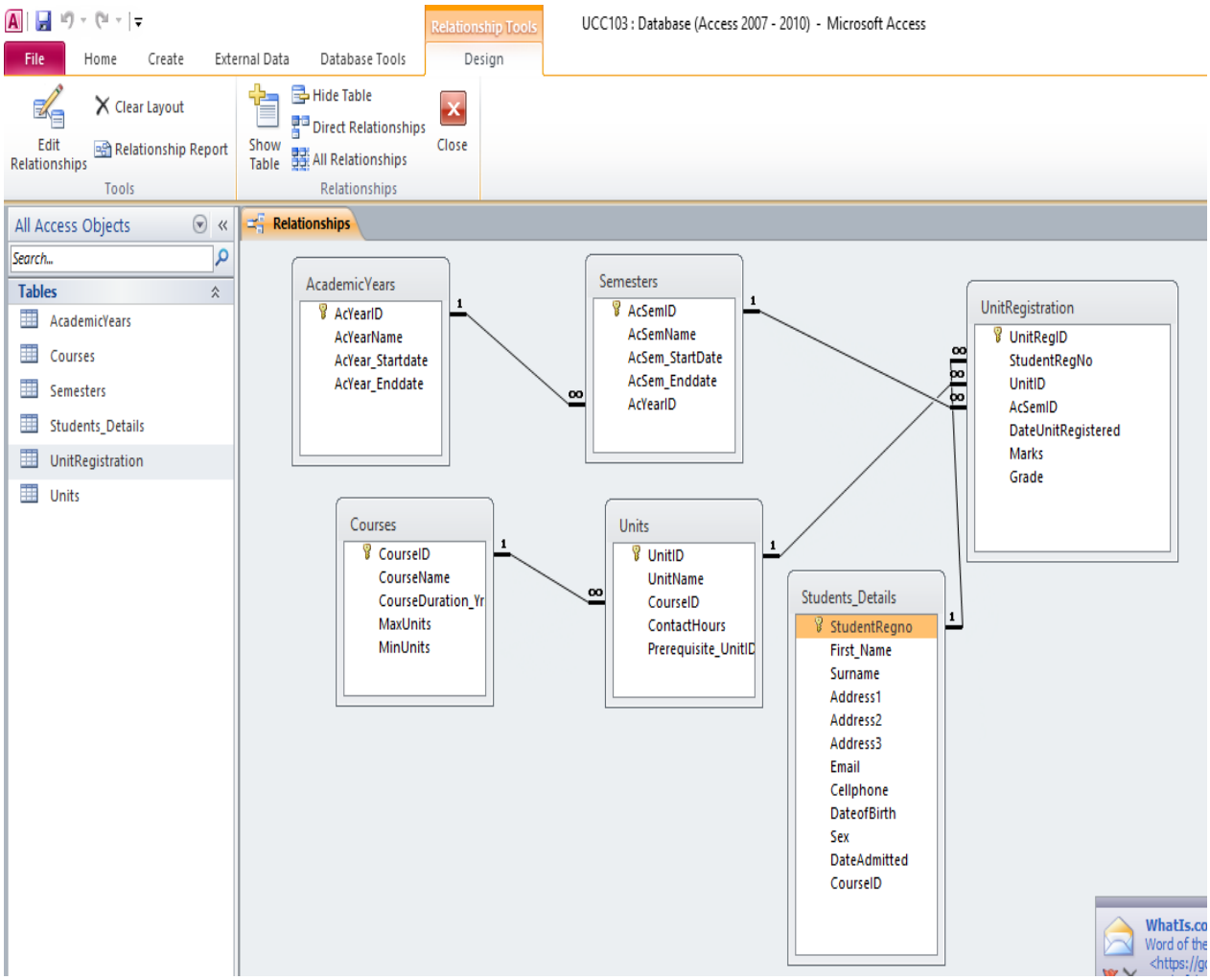
3) Select each of the list tables and Click Add Button
4) Close the Show tables window

On the Relationships Window

1) Select the one of the field and while holding the Mouse button Drag the pointer and click the related field, release the button
2) In the Edit relationships window
3) Check the Enforce referential integrity chech-box
4) Click Create button



5) Create the Database Relationships as shown below.



6) Click Save button
7) Click close button to close the window

12.10. Working with Queries

Query is a database search. It enables you retrieve data that meets certain criteria ie filter data from a database. We can use a query to

1) Create a report

2) Create a Table
3) Create a form

Types of Queries;

1) **Select query**: Select queries are similar to filters in that you can use either to:
    a. Retrieve a subset of records from an underlying table or query.
    b. Produce results that can be used as the source of data for a form or report.
    c. Sort records.
    d. Edit data if editing is allowed.
2) **Update query**: This is used when you want to make changes to a group of records in one or more tables all at once. It changes all specified records at once to reflect the correct information For example: when you wish to raise the fee of an amount paid in your database by 10%.
3) **Delete query**: A delete query removes records you specify e.g. you can remove outdated records
4) **Make table Query:** used to create a table based on the data defined by the query
5) **Parameter query**: uses dialog boxes that prompts for information, such as criteria for retrieving records or a value you want to insert in a field.

When you have a frequently asked query, you can save it so that each time you need the information, you just run the query instead of opening the table.

For Queries, we shall use the Northwind Sample Database

Installing the Northwind Database

Access database templates used to be downloadable from the web, but they are now available only from within Access itself. Here's how to install the Northwind database in Microsoft Access 2010.

1) Open Microsoft Access 2010.
2) On the New tab (opened by default when you start Access), select **Sample Templates** under the **Available Templates** section.
3) Click on **Northwind**. This may also be called Northwind 2007 Sample.
4) In the **File Name** text box on the right side of the screen, provide a filename for your Northwind database.
5) Click the **Create** button. Access downloads the Northwind database from Microsoft and prepares your copy. This may take a few minutes.
6) The database opens automatically when it has completed downloading.

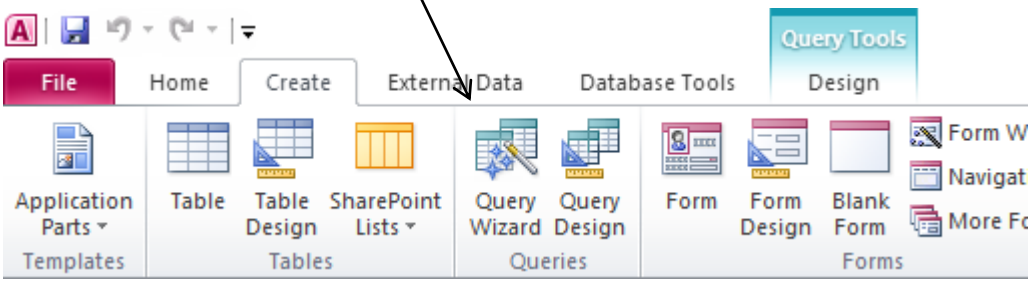
12.10.1. Creating a query

You can create queries using either:-

- In design view
- Using wizard

1) Select the Create menu

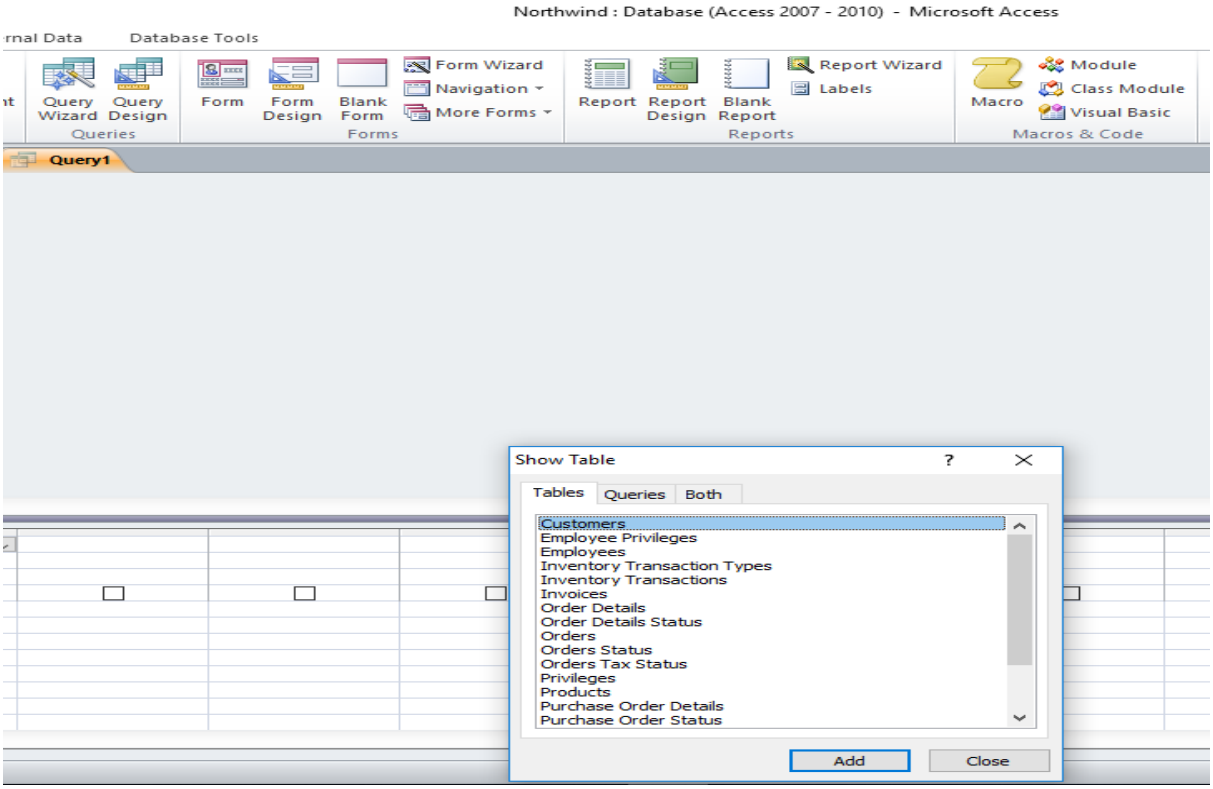
2) Under the Queries section Select the appropriate method



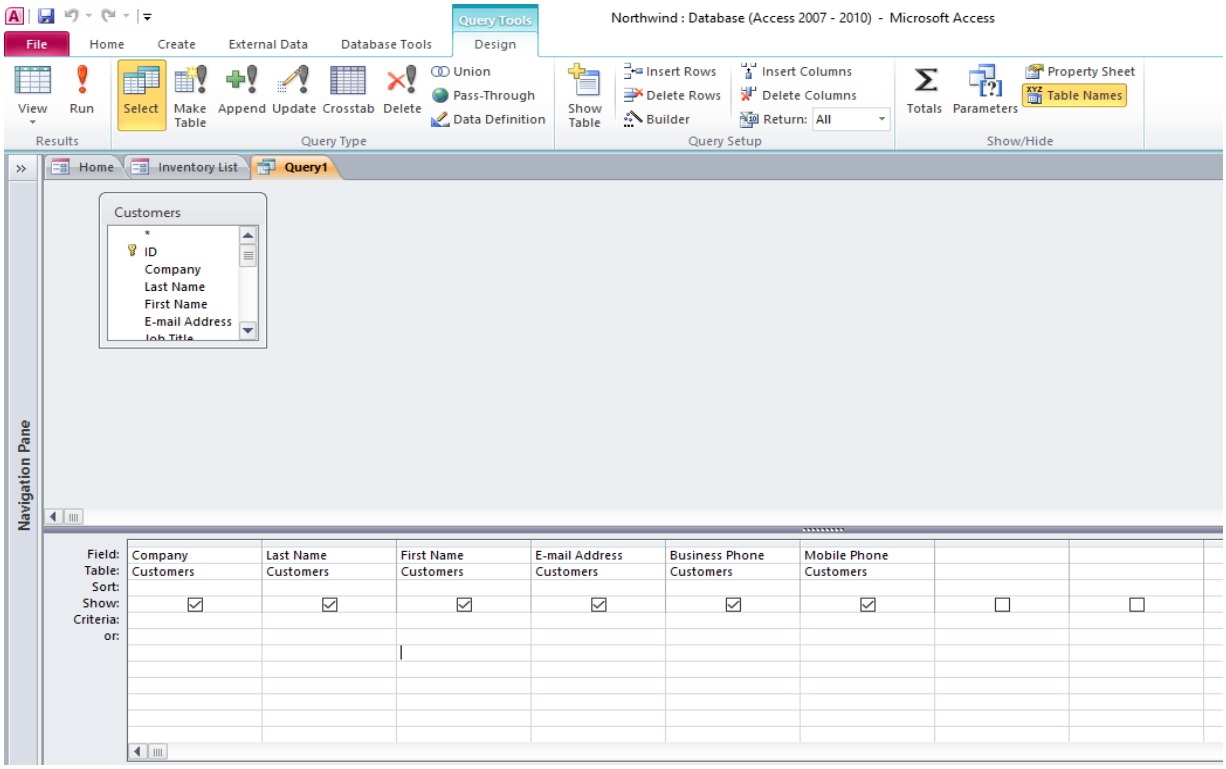
Creating a select query

To show a list of cutomers in the Northwind database

1) Click Query Design

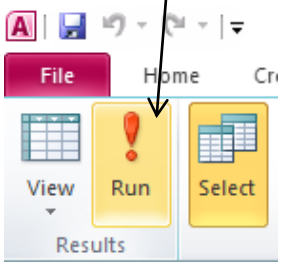


2) Select the Customer table (Select to highlight the table from the table list, then click Add button).
3) Click close to close the show table window
4) Select the details of a customer required in the query by selecting the field and double clicking. The selected field will appear in the grind at the bottom of the page as show below;

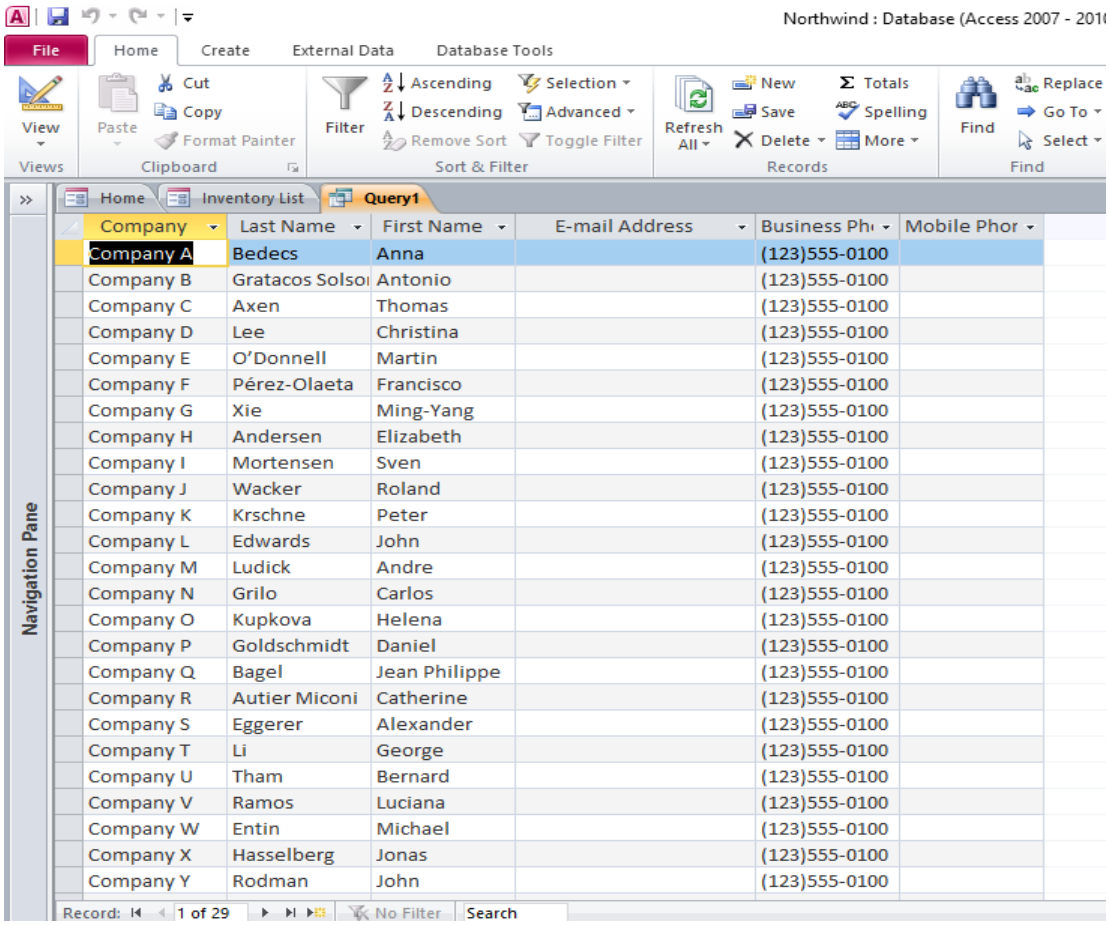


12.10.2. To Run the Query

Click the Run Command in the results section of the menu



Results – a list of the customer records in the database



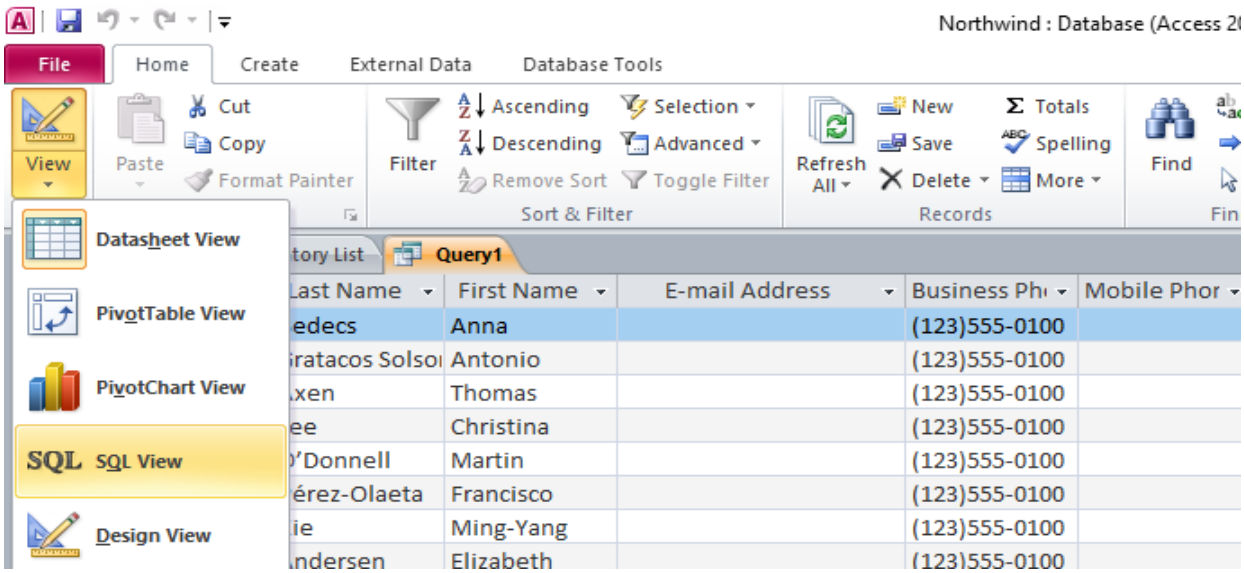
12.10.3. SQL Queries

An SQL query is a query you create by using a Structured Query Language (SQL) statement. You can use SQL to query, update, and manage relational databases such as Access.

When you create a query, Access automatically constructs the equivalent SQL statements for you. If you want, you can view or edit the SQL statement in SQL view. However, after you make changes to a query in SQL view, the query might not be displayed the way it previously was in Design view.

To view the SQL statement created for the Query above

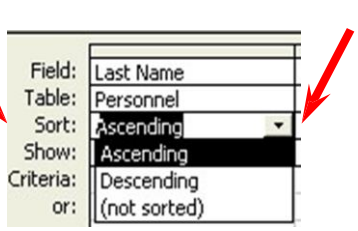
1) Click View menu
2) Select SQL Sql View



The SQL statement is;

SELECT Customers.Company, Customers.[Last Name], Customers.[First Name], Customers.[E-mail Address], Customers.[Business Phone], Customers.[Mobile Phone]
FROM Customers;

12.11. Sorting the Database



If you are not in the Query Design Screen, you'll need to be in that view. So, go to the Design Screen. Notice that the third row in the lower half of the screen indicates Sort: (like the image at the right). Click-in the Sort: area under Last Name. A down triangle box appears; click-on the down triangle. Let's sort the Last Names in Ascending order. Click-on Ascending. Notice that Ascending now appears in the Sort: area. Click-on the (!) to see the new query. Notice that the names you entered are alphabetized. Click-on the Design View button (triangle-ruler-pencil). Now change the Ascending under Last Name to (not sorted). On your own, try sorting some of the other fields. When you are finished remember to set the fields to (not sorted) unless you do want to sort on those fields.

12.12. Forms

Working with forms

A form is basically a way of displaying data, record by record. It is the most convenient object in which a non-experienced user of Access can enter records. Any record entered using a form is automatically entered into the table. In forms, you can edit, sort, filter, add/delete records etc.

In addition, you can customize a form to view and print information in a style that you like you can even add pictures to make the form more visually appealing. If you want to organize fields

into subcategories, you can create a multipage form.

To create a form using wizard

1) Select the table or Query the form will be based
2) On the Create Menu, click Form Icon
3) A Form structure appear in Design mode. You can make any modification of the form design such as resizing the fields, renaming the field names etc at this stage.
4) To Save the form, Select Save As and provide the name of the form.

Use form for Data Entry

a) Click View Icon
b) Enter the relevant details in each field and press < Enter> at the end of each entry.

12.13. Reports

Reports are used to analyze data or present it in a certain way in print. A report could be that which groups data and calculates totals and another different data formatted for printing mailing tables.

To create A Report using Wizard

You can create a report on your own or you can have Ms-Access create one for you .The wizard speeds up the process since it does all the basic work for you. A wizard promotes you for information and creates a report based on your answers. You can also customize a table in design view.

Procedures

1) Click report wizard icon on the create Menu
2) Select the table or query the report will be based on, and click next
3) Select the fields to be displayed in the report, and click next
4) Select the field or fields to be used for grouping the data and click next
5) Select the fields ordering format (Ascending or Descending), and click next
6) Select the report format required, and click next
7) Select the report style, and click next.
8) Give the report title
9) Click finish

12.14. Chapter Review Exercise

Prepare a database table and assign appropriate data types to hold the following data and define a primary key

Name	**Course**	**DATE OF BITH**	**SEX**	**FEES(Kshs.)**
Kilonzo David Muthami	B. Ed	31/01/92	Male	54,000
Mumbi Irene Wangui	M. Ed	06/10/87	Female	60,000
Mwangi Muriithi	M. Ed	21/06/93	Male	54,000
Simon Kinyanjui	BBIT	16/09/97	Male	62,000
Muuki Agneta Kalekye	B. Ed	19/04/93	Female	60,000

Wang'ombe Grace	BBIT	17/01/88	Female	54,000
Kiramana Paul	M. Ed	25/04/90	Male	19,000
Gatwiri Everlyn	B. Ed	11/11/92	Female	60,000
Ruth Wangeci	BBIT	31/01/92	Female	16,000
Harrison Mundia	B. Ed	2/01/89	Male	22,000
Musyoka Ruth	B. Ed	31/01/92	Female	36,000
Mwenda Sylivia	BBIT	19/08/94	Female	41,000
Reuben Gichana	M. Ed	11/05/92	Male	56,000
Mutio Mary Mwikali	B. Ed	31/12/91	Female	60,000
Karanja Kelvin	BBM	26/11/92	Male	60,000

Required

1. Create a data entry form in design view and apply your own design in the form
2. Enter the above records using the form above
3. The semesters fees is 60,000 Kshs. Produce a list of students who have cleared the fees
4. Produce a list of Female students taking a course in education
5. Produce a list of students who were born before the year 1990
6. Product a report that will show the current students grouping them by their respective courses
7. Prepare a report showing a list of students who have cleared their fees
8. Prepare a macro that will open no 3 above and open the report on 7 above.